

TURNING POINT

FAIL-SAFE GUNS

Matchlocks, wheel-locks, and flintlocks used a small amount of gunpowder to prime the propellant (main gunpowder charge). In 1807, the Reverend Alexander Forsyth patented a way of igniting the propellant by using a different substance—a sensitive chemical primer that detonates when struck. Joshua Shaw later patented the percussion cap as the simplest way of making Forsyth's invention work. Firearms could now use chemical ignition. This key development in firearms technology enabled guns to fire instantaneously and reliably, unlike earlier guns with exposed gunpowder priming. It also enabled the development of the revolver and the self-contained metallic cartridge (see pp. 122–23), now used by nearly every modern firearm.



▲ PERCUSSION CAPS

Percussion caps were small copper or brass cups containing a minute quantity of fulminate. A cap was held in place on a hollow plug, or nipple, that was attached to the breech of the gun.

In the early 19th century, Alexander Forsyth, an avid duck hunter, was frustrated by the shortcomings of the flintlock system. Although reliable, it suffered from the occasional “flash in the pan” when the priming powder would ignite but the gun would fail to fire. Along with the noise of the flint striking the frizzen and the puff of smoke, the “flash” alerted potential game, which would quickly disappear.

» BEFORE

At the beginning of the 19th century, most guns were fired by the flintlock mechanism. In this, a piece of flint was struck against steel to create sparks that ignited some priming powder in a small pan alongside the barrel. The flame from this passed through a vent in the barrel and ignited the main charge.

• **LOOSE POWDER PLACED IN A PRIMING PAN** in small quantities was not efficient. Wind could blow it away and rain could make it wet. The powder could also ignite but fail to detonate the main charge.

• **DELAYS BETWEEN PULLING THE TRIGGER** and the gun actually discharging gave time for birds and animals, startled by the flash and smoke of the ignited priming powder, to escape.



FLINTLOCK MECHANISM

• **FLINTS NEEDED TO BE REPLACED** after 15 shots or so, and the quality of flints often varied. The hard steel face of the frizzen also wore out, reducing its ability to create a spark.

THE “SCENT-BOTTLE” LOCK

Forsyth set about devising a simpler, faster, and more effective means of ignition. He designed a mechanism that could be attached to any firearm. It used a detonating compound called mercury fulminate as a primer to ignite the main powder charge. The fulminate was held in a vessel shaped like a perfume bottle, which gave this mechanism the name “scent-bottle” lock. It was mounted on a hollow, cylindrical spindle and screwed into a flintlock gun's vent that had been specially enlarged.

Forsyth's invention embodied the fundamental principles of chemical ignition upon which all future gun and ammunition development would be based.

PERCUSSION DESIGN EVOLVES

Although revolutionary, the “scent-bottle” lock was unsafe as it carried a large quantity of a detonating compound, which could explode accidentally and injure the user. Many people attempted to adapt Forsyth's idea to design a variety of safer percussion systems that would use a tiny, isolated quantity of primer—just enough to prime the gun once. The gunmaker

▼ THE THIN RED LINE

Armed mainly with Pattern 1851 percussion rifles, the 93rd Highlanders regiment of the British Army bravely formed an unmoving line of defense against the Russian cavalry in the Battle of Balaclava in 1854. From a distance, they appeared to onlookers as a “thin red line” because of their red coats.

